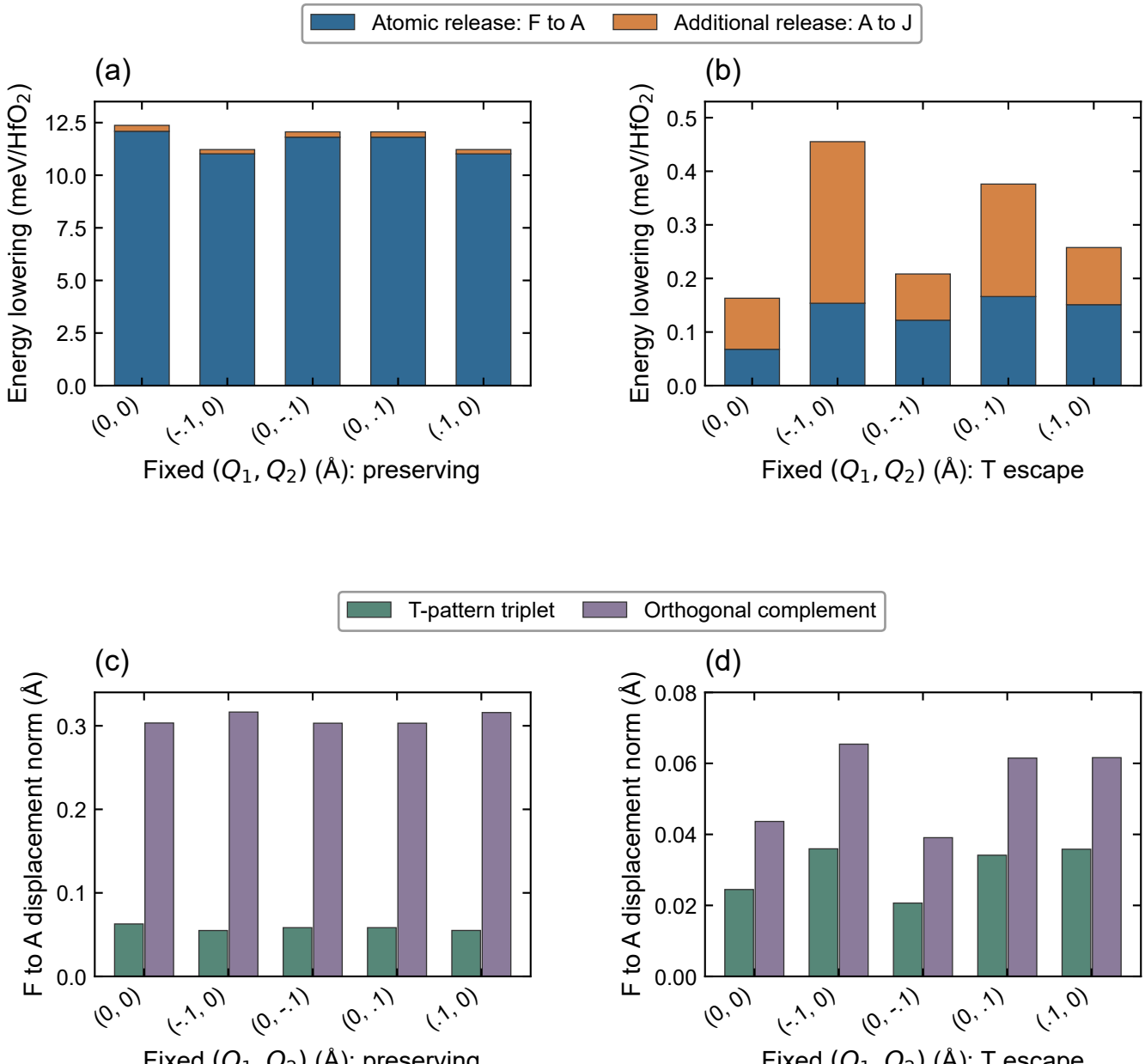
Atomic release: F to A
Additional release: A to J
(a)
Energy lowering (meV/HfO2)
12.5
10.0
7.5
5.0
2.5
0.0
(0, 0)
(-1, 0)
(0, -1)
(0, 1)
(1, 0)
Fixed (Q1, Q2) (Å): preserving
(b)
Energy lowering (meV/HfO2)
0.5
0.4
0.3
0.2
0.1
0.0
(0, 0)
(-1, 0)
(0, -1)
(0, 1)
(1, 0)
Fixed (Q1, Q2) (Å): T escape
T-pattern triplet
Orthogonal complement
(c)
F to A displacement norm (Å)
0.3
0.2
0.1
0.0
(0, 0)
(-1, 0)
(0, -1)
(0, 1)
(1, 0)
Fixed (Q1, Q2) (Å): preserving
(d)
F to A displacement norm (Å)
0.08
0.06
0.04
0.02
0.00
(0, 0)
(-1, 0)
(0, -1)
(0, 1)
(1, 0)
Fixed (Q1, Q2) (Å): T escape